

THE WEIL QUADRATIC FORM OF THE CONNES–CONSANI–MOSCOVICI PROLATE VECTOR, AND A μ^8 UPPER BOUND FOR WINDOWED WEIL FORMS

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ABSTRACT. Weil’s criterion states that the Riemann Hypothesis holds if and only if Weil’s quadratic form W is nonnegative. Restricted to test functions supported in a multiplicative window $[\lambda^{-1}, \lambda]$, this becomes $\lambda_{\min}(\lambda) \geq 0$ for all λ , where $\lambda_{\min}(\lambda)$ is the bottom of the spectrum of the restricted form; any proof of RH along these lines must control how fast this margin shrinks. Put $\mu = \lambda^2$. In earlier work we proved, without any hypothesis on the zeros of ζ , that $\lambda_{\min}(\lambda) \leq 1.362 \times 10^9 \mu^{23} e^{-4\pi\mu}$ for $\mu \geq 5$. Connes, Consani and Moscovici proposed a vector k_λ built from two prolate spheroidal wave functions, and observed numerically that $W(k_\lambda)$ tracks the prolate eigenvalue defect $1 - \chi_2$. We prove this unconditionally. Theorem C bounds $|W(k_\lambda)|$ by an explicit polynomial in μ times the out-of-band mass $\rho_{\text{out}} \leq 1 - \chi_2^2$ of the underlying band-limited function, plus $e^{-16\mu}$. We then show that $\|k_\lambda\|^2 \geq 0.109 \|h_\lambda\|^2$ for $\mu \geq 50$ by comparison with Hermite functions. Combined with a non-asymptotic bound $1 - \chi_2^2 \leq C\mu^{9/2} e^{-4\pi\mu}$ from a companion paper, this gives

$$\lambda_{\min}(\lambda) \leq 5.011 \times 10^{17} \mu^8 (\log \mu)^3 e^{-4\pi\mu} \quad (\mu \geq 50).$$

All constants are explicit and established with ball arithmetic. The results are upper bounds only and say nothing about positivity.

Status. Proved in this paper, with explicit constants established by ball arithmetic (Arb). Not formalized. This paper has not yet been reviewed by independent human experts.

1. INTRODUCTION

1.1. Background. Let W be Weil’s quadratic form and W_λ its restriction to test functions supported in the multiplicative window $[\lambda^{-1}, \lambda]$ [1, 2]. Write $\lambda_{\min}(\lambda)$ for the bottom of its spectrum and $\mu = \lambda^2$. By Weil’s criterion, RH is equivalent to $\lambda_{\min}(\lambda) \geq 0$ for all λ .

In [5] we proved two unconditional upper bounds:

$$\lambda_{\min}(\lambda) \leq 1.563 \times 10^9 \mu^8 e^{-2\pi\mu} \quad \text{and} \quad \lambda_{\min}(\lambda) \leq 1.362 \times 10^9 \mu^{23} e^{-4\pi\mu} \quad (\mu \geq 5).$$

[5] left two questions open.

- *Remark 3 there:* does $W(k_\lambda)$ admit a bound by a multiple of the prolate eigenvalue defect $1 - \chi_2$? Here k_λ is the vector of Connes, Consani and Moscovici [2, §7]. Numerically $W(k_\lambda)/\|k_\lambda\|^2 \approx (8\text{--}15)(1 - \chi_2)$.
- *Remark 2 there:* the power μ^{23} was not optimized.

This paper answers the first question and reduces the power to μ^8 up to logarithms.

1.2. Results. Let ψ_n be the prolate spheroidal wave functions of bandwidth $c = 2\pi\mu$, and $h_{n,\lambda}(x) = \psi_n(x/\lambda)/\sqrt{\lambda}$ on $|x| < \lambda$. Put

$$h_\lambda = I_4 h_{0,\lambda} - I_0 h_{4,\lambda} \quad (I_n = \int h_{n,\lambda}), \quad k_\lambda = \mathcal{E}(h_\lambda)|_{[\lambda^{-1}, \lambda]}.$$

Let χ_0, χ_2 be the eigenvalues of the finite Fourier transform on $h_{0,\lambda}, h_{4,\lambda}$, and

$$\rho_{\text{out}} = \|\hat{h}_\lambda \mathbf{1}_{|\xi| > \lambda}\|^2 / \|h_\lambda\|^2 = \frac{I_4^2(1 - \chi_0^2) + I_0^2(1 - \chi_2^2)}{I_0^2 + I_4^2} \leq 1 - \chi_2^2.$$

Theorem C. For every $\mu \geq 5$, without any hypothesis on the zeros of ζ ,

$$|W(k_\lambda)| \leq P(\mu) \rho_{\text{out}} \|h_\lambda\|^2 + e^{-16\mu} \|h_\lambda\|^2,$$

where $P(\mu)$ is the explicit function (6.1). It satisfies $P(\mu) = O(\mu^{7/2}(\log \mu)^3)$ and $P(5) \leq 1.854 \times 10^{13}$.

Theorem D. For every $\mu \geq 50$, $\|k_\lambda\|^2 \geq 0.109 \|h_\lambda\|^2$; for every $\mu \geq 80$, $\|k_\lambda\|^2 \geq 0.161 \|h_\lambda\|^2$. Moreover $\liminf_{\mu \rightarrow \infty} \|k_\lambda\|^2 / \|h_\lambda\|^2 \geq \kappa_\infty$, where $\kappa_\infty = 0.2192471995 \dots$ is an explicit Hermite constant (numerically the ratio converges to κ_∞).

Theorem E. For every $\mu \geq 50$,

$$\lambda_{\min}(\lambda) \leq 5.011 \times 10^{17} \mu^8 (\log \mu)^3 e^{-4\pi\mu}.$$

Theorem E uses the bound $1 - \chi_2^2 = 1 - \lambda_4(2\pi\mu) \leq C_4(2\pi\mu)^{9/2} e^{-4\pi\mu}$ from the companion paper [4]. The power 9/2 is that of Fuchs' asymptotic formula [3]. Compared with [5, Thm. B], the bound of Theorem E is smaller by a factor of about 1.4×10^{15} at $\mu = 50$, 2.8×10^{19} at $\mu = 100$, and 8×10^{33} at $\mu = 1000$.

1.3. Numerical context.

- The observed ratio $W(k_\lambda)/(\rho_{\text{out}} \|h_\lambda\|^2)$ is 1.6–2.3 for $5 \leq \mu \leq 15$, so the factor $P(\mu)$ in Theorem C is far from sharp. The loss comes mainly from the zero side, namely the zero-free region at height $e^{20\mu}$.
- The ratio $\|k_\lambda\|^2 / \|h_\lambda\|^2$ is about 0.22 for all $\mu \geq 5$.

1.4. Method.

- **Theorem C.** As in [5], $\mathcal{E}(h_\lambda)$ lies in the radical of W , so $W(k_\lambda) = \sum_\rho m_\rho \hat{r}(\gamma_\rho) \hat{r}(-\gamma_\rho)$, where r is the part of $\mathcal{E}(h_\lambda)$ outside the window. New ingredients:
 - (1) a Poisson representation of r , in which the oscillating sum $\sum \sin(mx + \Phi)/m$ is summed exactly;
 - (2) amplitude and phase estimates for the band-limited extension of ψ_n , from Liouville–Green theory and an endpoint estimate $\psi_n(1)^2 \leq 4c(1 - \lambda_n)/\lambda_n$;
 - (3) a disc-averaging bound for the zero sum, using subharmonicity, the zero-free region of Mossinghoff–Trudgian and the verification of RH to $3 \cdot 10^{12}$.
- **Theorem D.** The rescaled prolate operator converges to the harmonic oscillator. Hermite residuals of size $O(1/\mu)$, an IMS gap estimate and a Davis–Kahan argument in the form norm give weighted $O(1/\mu)$ control. A lattice-sum bound then transfers it to k_λ .

2. SETTING

2.1. Conventions. A word on notation: r denotes the part of e outside the window (Section 2); in Section 3 and Appendix B the letter r inside formulas in t stands for $t^2 - s$; r_d , r_T , r_ρ are disc radii in Section 5. Similarly ρ denotes a zero of ζ , ρ_{out} the out-of-band mass, and $\rho_n(t)$ a phase remainder; the Hermite residual norms are ϱ_n . Throughout, $\lambda > 1$, $\mu = \lambda^2 \geq 5$, $a = \log \lambda$ and $c = 2\pi\mu \geq 10\pi$. Fourier transforms are $\hat{f}(z) = \int f(y) e^{izy} dy$ in the logarithmic variable and $\hat{\varphi}(\xi) = \int \varphi(x) e^{2\pi i x \xi} dx$ in the additive variable, as in [5]. For a nontrivial zero ρ of multiplicity m_ρ put $\gamma_\rho = (\rho - \frac{1}{2})/i$. Thus $\text{Re } \gamma_\rho = \text{Im } \rho$ is the ordinate, and $|\text{Im } \gamma_\rho| = |\text{Re } \rho - \frac{1}{2}|$.

2.2. Prolate functions.

- ψ_n is the n -th prolate spheroidal wave function of bandwidth c , normalized in $L^2(-1, 1)$. It is an eigenfunction of time and frequency limiting with eigenvalue $\lambda_n = \lambda_n(c)$, and of $L_c = -\frac{d}{ds}(1 - s^2)\frac{d}{ds} + c^2 s^2$ with eigenvalue χ_n^{SL} .
- $\tilde{\psi}_n$ is its band-limited extension to $\mathbb{R}$. It satisfies $\|\tilde{\psi}_n\|_{L^2(\mathbb{R})}^2 = 1/\lambda_n$ and $\int_{|t|>1} \tilde{\psi}_n^2 = (1 - \lambda_n)/\lambda_n$.

- For $n \in \{0, 4\}$ we write $\chi_{(n)} = \lambda_n^{1/2}$, so $\chi_0 = \chi_{(0)}$ and $\chi_2 = \chi_{(4)}$ in the notation of [2, 5].

2.3. The vector. Put $h_{n,\lambda}(x) = \psi_n(x/\lambda)/\sqrt{\lambda}$ on $|x| < \lambda$ (and 0 elsewhere), $I_n = \int h_{n,\lambda}$, $\alpha = I_4$, $\beta = -I_0$ and $h = \alpha h_{0,\lambda} + \beta h_{4,\lambda}$. Then $\int h = 0$ and $\|h\|^2 = \alpha^2 + \beta^2$. Further,

$$e(y) = e^{y/2} \sum_{n \geq 1} h(ne^y), \quad k = e \mathbf{1}_{[-a,a]} = k\lambda, \quad r = e \mathbf{1}_{(-\infty, -a)}.$$

In Fourier space $\hat{h}(\lambda t) = \sum_n w_n \tilde{\psi}_n(t)$, with $w_n = \text{coef}_n \chi_{(n)}/\sqrt{\lambda}$, $\text{coef}_0 = \alpha$ and $\text{coef}_4 = \beta$. Put

$$\Psi = |\alpha| \chi_0 |\psi_0(1)| + |\beta| \chi_2 |\psi_4(1)|.$$

Lemma 2.1 (out-of-band identity). $\rho_{\text{out}} = (\alpha^2(1 - \lambda_0) + \beta^2(1 - \lambda_4))/(\alpha^2 + \beta^2)$.

Proof. By Plancherel, $\|\hat{h}_{n,\lambda}\| = 1$. The in-band part of $\hat{h}_{n,\lambda}$ is $\chi_{(n)} h_{n,\lambda}$, and $h_{0,\lambda} \perp h_{4,\lambda}$. Hence the out-of-band cross term is $\langle \hat{h}_0, \hat{h}_4 \rangle - \chi_0 \chi_2 \langle h_0, h_4 \rangle = 0$. $\square$

2.4. The radical lemma and the explicit formula.

Lemma 2.2. (a) $|e(y)| \leq \text{TV}(h)e^{y/2}$ for $y \leq -a$.

(b) $\hat{r}$ is holomorphic on $\text{Im } z < \frac{1}{2}$, and $\hat{k}(\pm\gamma_\rho) = -\hat{r}(\pm\gamma_\rho)$ for every nontrivial zero.

(c) For $\tau < \frac{1}{2}$, $\int |\hat{r}(s + i\tau)|^2 ds = 2\pi \|r_\tau\|^2$, where $r_\tau(X) = X^\tau r(-\log X)$ in $L^2((\lambda, \infty), dX/X)$.

Proof. (a) For $x = e^y \leq 1/\lambda$, $x \sum h(nx)$ is a Riemann sum for $\int_0^\infty h = 0$, with error at most $x \text{TV}(h)$.

(b) For $\text{Im } z < -\frac{1}{2}$, Fubini gives $\hat{e}(z) = \zeta(\frac{1}{2} + iz) \int_0^\lambda h(v) v^{-1/2+iz} dv$. The integral is holomorphic on $\text{Im } z < \frac{1}{2}$, so the identity extends there, minus the pole. Since $e = k + r$ and $\hat{e}(\pm\gamma_\rho) = 0$, the claim follows.

(c) This is Plancherel. $\square$

By the explicit formula [5, Lemma 3.2] and Lemma 2.2(b),

$$(2.1) \quad W(k) = \sum_{\rho} m_{\rho} \hat{r}(\gamma_{\rho}) \hat{r}(-\gamma_{\rho}), \quad |W(k)| \leq \sum_{\rho} m_{\rho} |\hat{r}(\gamma_{\rho})|^2,$$

where the last step is Cauchy–Schwarz together with the symmetry $\gamma \mapsto -\gamma$ of the zeros.

Lemma 2.3 (Poisson form). For a.e. $X > \lambda$, $r(-\log X) = X^{1/2} \sum_{m \geq 1} \hat{h}(mX) - \frac{1}{2} h(0) X^{-1/2}$.

Proof. h is even and of bounded variation, with jumps only at $\pm\lambda$. Apply the Dirichlet–Jordan test to the periodization, using symmetric partial sums and $\hat{h}(0) = 0$. $\square$

3. THE PROLATE SIDE

Lemma 3.1 (eigenvalues and variation). For $c \geq 10\pi$ and $n \leq 4$:

- $\lambda_n \geq (1 - 5\varepsilon_*)/(1 - \varepsilon_*) > 395/399$, where $\varepsilon_* = 16 \cdot 14!/30^{10}$;
- $|\psi_n| \leq \sqrt{c/(\pi\lambda_n)}$ and $|\psi'_n| \leq c^{3/2}/\sqrt{3\pi\lambda_n}$ on $[-1, 1]$;
- the total variation V of k on $\mathbb{R}$, including the jumps at $\pm a$, satisfies $V \leq 9\mu^3 \|h\|$.

Proof. • **Eigenvalues.** Apply min–max to Q_c on the span of the five scaled Hermite functions $c^{1/4} \tilde{H}_k(\sqrt{c}s)$. Their time and band leakage is at most ε_* .

• **Pointwise bounds.** These follow from $\psi_n = \mu_n^{-1} \mathcal{F}_c \psi_n$ and Cauchy–Schwarz.

• **Variation.** At most μ terms of e contribute in the window. Each has total variation at most $3\sqrt{\lambda}M + \lambda^{3/2}D$, where $M = \sup |h|$ and $D = \sup |h'|$ are bounded via the pointwise bounds.

Full details are in Appendix A. $\square$

Lemma 3.2 (Liouville–Green). Let $c \geq 10\pi$, $n \in \{0, 4\}$, $s = \chi_n^{\text{SL}}/c^2$, $p = t^2 - 1$ and $q = c^2(t^2 - s)$.

- (a) $s < 0.3$.
- (b) $V(t) = \tilde{\psi}^2 + (p\tilde{\psi}')^2/(pq)$ is non-increasing on $t > 1$, so $|\tilde{\psi}_n(t)| \leq |\psi_n(1)|$ for $|t| \geq 1$.
- (c) With $z' = \sqrt{q/p}$ and $\tilde{\psi} = (pq)^{-1/4}w$, we have $w_{zz} + (1 + \delta)w = 0$, where

$$\delta = \frac{1}{4c^2pr} - \frac{s^2 + 2s + (3 - 6s)t^2}{4c^2r^3}, \quad r = t^2 - s,$$

and $|\delta|c^2t^4 \leq 0.5038$ for $t \geq 2$.

Proof. (a) χ/c^2 is non-increasing in c by Hellmann–Feynman. At $c = 10\pi$, a Sturm count in ball arithmetic shows that the third even eigenvalue is below $0.3c^2$.

(b) We have $V' = -(p\tilde{\psi}')^2(pq)'/(pq)^2 \leq 0$ and $V(1^+) = \psi(1)^2$.

(c) The formula is direct differentiation. The bound on δ is a ball-arithmetic bound over $u = t^{-2} \in [0, \frac{1}{4}]$ and $s \in [0, \frac{1}{2}]$. $\square$

Lemma 3.3 (endpoint value). *For $n \in \{0, 4\}$ and $c \geq 10\pi$, $\psi_n(1)^2 \leq 4c(1 - \lambda_n)/\lambda_n$. Consequently $\Psi^2 \leq 8c\rho_{\text{out}}\|h\|^2$.*

Proof. Let $A_0 = |\psi_n(1)|$, change the sign of ψ_n so that $\psi_n(1) = A_0$, and let $t_0 = 1 + 1/(2c^2)$.

- (1) Integrating $(p\tilde{\psi}')' + q\tilde{\psi} = 0$ once gives $|p\tilde{\psi}'| \leq A_0c^2(t-1)t^2$. Hence $\tilde{\psi}(t_0) \geq \frac{37}{50}A_0$.
- (2) The Liouville–Green error satisfies $J := \int_{t_0}^{\infty} |\delta| dz \leq \frac{339}{980} < \frac{7}{20}$.
- (3) By Gronwall, the energy $E = w^2 + w_z^2$ stays within $[\frac{7}{10}, \frac{10}{7}]E(t_0)$ on $[t_0, \infty)$.
- (4) A weighted energy identity on $[4/3, \infty)$ gives $2 \int_{4/3}^{\infty} \tilde{\psi}^2 \geq A_0^2/(4c)$.
- (5) The mass identity $\int_{|t|>1} \tilde{\psi}^2 = (1 - \lambda_n)/\lambda_n$ then gives the claim.

The consequence follows from Cauchy–Schwarz and Lemma 2.1. Full details are in Appendix B. $\square$

Lemma 3.4 ($h(0)$). $h(0)^2 \leq 2.0408 \lambda \rho_{\text{out}}\|h\|^2$.

Proof. The eigenrelation at the centre gives $h(0) = I_0I_4(\chi_2 - \chi_0)/(\chi_0\chi_2)$. We have $(\chi_0 - \chi_2)^2 \leq 1 - \chi_2^2$ and $I_4^2 \leq 2\lambda$. Moreover $\lambda_0\lambda_4 \geq (395/399)^2$, and $2(399/395)^2 \leq 2.0408$. $\square$

Lemma 3.5 (amplitude). *Let $A_0 = |\psi_n(1)|$ and $E = w^2 + w_z^2$. Then:*

- $E \leq 3.12A_0^2$ on $t \geq t_0$;
- the limit $A_{\infty} = \lim E^{1/2}$ satisfies $A_{\infty} \leq 1.78A_0$;
- $\tilde{\psi}_n(t)^2 \leq 3.79A_0^2/(c\sqrt{t^2 - 1})$ for $1 < t \leq 2$.

Proof. **Initial energy.** From $|p\tilde{\psi}'| \leq A_0c^2(t-1)t^2$ and $|\tilde{\psi}| \leq A_0$, at t_0 :

- $w^2 \leq t_0(1 + \frac{1}{4c^2})^{1/2}A_0^2$;
- $|w_z| \leq A_0[\frac{1}{2}t_0^2Z^{-1/4} + \frac{1}{2}t_0(Z^{-1/4} + (c^2p)^{3/4}r_{\text{lo}}^{-5/4}/c^2)]$, where $r_{\text{lo}} = t_0^2 - 0.3$ and $Z = c^2p(t_0)r_{\text{lo}}$ (so $Z \geq 0.7$ for all c , and $Z > 0.7011$ at $c = 10\pi$).

Every term decreases in c . At $c = 10\pi$, ball arithmetic gives $E(t_0) \leq 2.1984A_0^2$.

Propagation. By Gronwall with $J \leq \frac{7}{20}$, $E \leq e^{7/20}E(t_0) \leq 3.1197A_0^2$.

Limit and near field. $\log E$ has an integrable derivative, so the limit exists. For $t \geq t_0$, $\tilde{\psi}^2 = (pq)^{-1/2}w^2$ and $pq \geq 0.7c^2(t^2 - 1)$ give the near-field bound. For $1 < t < t_0$, use $|\tilde{\psi}| \leq A_0$ and $3.79/(c\sqrt{t^2 - 1}) > 1$. $\square$

Lemma 3.6 (far field). *For $t \geq 2$, $\tilde{\psi}_n(t) = A_{\infty}(ct^2)^{-1/2}[\sin(ct + \Phi_n) + \rho_n(t)]$, where $|\rho_n(t)| \leq \min(2, 2c/(3t)) + 0.4/t^2$.*

Proof. Use the Prüfer variables $w = A \sin \theta$, $w_z = A \cos \theta$. We have $\int_t^{\infty} |\delta| dz \leq 0.2909/(ct^2)$. Hence:

- the phase error is $|\varepsilon_1| \leq 0.00926t^{-2}$;
- the amplitude error is $|\varepsilon_2| \leq 0.00464t^{-2}$;

- the normalizing factor is $(pq)^{-1/4} = (ct^2)^{-1/2}(1 + \eta)$ with $\eta \leq 0.3829t^{-2}$, by convexity;
- the geometric phase is $|z - ct - Z_0| \leq 2c/(3t)$.

□

4. THE REMAINDER

Lemma 4.1. For $0 \leq \tau < \frac{1}{2}$, $\|r_\tau\|^2 \leq \lambda^{2\tau}(1 - 2\tau)^{-1}M$, where

$$M \leq (592 + 152.2K_1^2) \rho_{\text{out}} \|h\|^2, \quad K_1 = 2 \log(c/3) + 8.08.$$

For $\tau < 0$, $\|r_\tau\| \leq \|r_0\|$.

Proof. Put $u = X/\lambda$ and $B_n = \lambda^{1/2} w_n A_{\infty, n} c^{-1/2}$. Then $B = |B_0| + |B_4| \leq 1.78\Psi/\sqrt{c}$ by Lemma 3.5. For $mu \geq 2$, Lemmas 2.3 and 3.6 give

$$X^{1/2} \hat{h}(mX) = u^{-1/2} \sum_n B_n m^{-1} [\sin(cmu + \Phi_n) + \rho_n(mu)].$$

- **Linear phase.** The sum is explicit: $\sum_m \sin(mx + \Phi)/m = -\sin \Phi \ell(x) + \cos \Phi (\pi - (x \bmod 2\pi))/2$, with $\ell(x) = \log |2 \sin(x/2)|$.
- **Phase errors.** $\sum_{mu \geq 2} |\rho_n(mu)|/m \leq 2 \log(c/3) + 5.49$.
- **The $m = 1$ term** for $1 < u < 2$ is bounded by Lemma 3.5.
- **Integration.** We use $\int_j^{j+1} \ell(cu)^2 du \leq \pi^2/12 + \pi^3/(3c) \leq 1.152$ and $\sum_j j^{2\tau-2} \leq 2/(1 - 2\tau)$.

This gives $M \leq (73.8 + 19.02K_1^2)\Psi^2/c + 0.75h(0)^2/\lambda$. Conclude with Lemmas 3.3 and 3.4. □

5. THE ZERO SUM

We use three explicit results.

- **Zero counting** [8, Cor. 1]. For $T \geq e$, $|N(T) - \frac{T}{2\pi} \log \frac{T}{2\pi e} - \frac{7}{8}| \leq R(\log T)$, where $R(x) = 0.111x + 0.275 \log x + 2.450 + 0.2/e$.
- **Zero-free region** [6, Thm. 1]. There are no zeros with $|t| \geq 2$ and $\sigma > 1 - 1/(5.573412 \log |t|)$.
- **Verification of RH** [7]. All zeros with $0 < \gamma \leq 3 \cdot 10^{12}$ lie on the critical line.

Lemma 5.1. Let $T \geq 3 \cdot 10^{12}$ and $r_T = 1/(11.15 \log(T + 1))$. Let n^* bound the number of zeros, with multiplicity, whose ordinate lies within $\frac{1}{4}$ of any s with $|s| \leq T + 1$. Then

$$|W(k)| \leq \frac{2n^*}{r_T^2} \left(\frac{1}{2} + \frac{\lambda}{2} \log \frac{1}{2r_T} \right) M + \frac{2\lambda V^2 (\log T + 1)}{\pi T}.$$

Proof. Start from (2.1) and put $F = \hat{r}$.

Discs. To each zero with $|\operatorname{Re} \gamma_\rho| \leq T$ attach the disc $D(\gamma_\rho, r_\rho)$.

- If $|\operatorname{Re} \gamma_\rho| \leq 3 \cdot 10^{12}$, take $r_\rho = \frac{1}{4}$; then $\operatorname{Im} \gamma_\rho = 0$.
- Otherwise take $r_\rho = 1/(11.15 \log |\operatorname{Re} \gamma_\rho|)$. The zero-free region gives $\frac{1}{2} - |\operatorname{Im} \gamma_\rho| \geq 2r_\rho$.

In both cases the discs lie in $|\operatorname{Im} z| \leq \frac{1}{2} - r_T$.

Low zeros. By subharmonicity, $|F(\gamma_\rho)|^2 \leq (\pi r_\rho^2)^{-1} \iint_{D_\rho} |F|^2$. Each point is covered at most n^* times. Lemmas 2.2(c) and 4.1 bound the resulting strip integral.

High zeros. Integration by parts gives $|\hat{k}(z)| \leq \lambda^{1/2} V / |\operatorname{Re} z|$. Moreover $\sum_{\gamma > T} m/\gamma^2 \leq (\log T + 1)/(\pi T)$, by $N(t) \leq \frac{t}{2\pi} \log \frac{t}{2\pi}$ for $t \geq T$. □

The zero count gives $n^* \leq N := (20\mu + 1 - \log 2\pi)/(4\pi) + 2R(20\mu + 1)$ for $T = e^{20\mu}$. The first zero has ordinate 14.13, so for $|s| < 13.9$ the count is at most $1 \leq N$.

6. PROOF OF THEOREM C

Take $T = e^{20\mu}$. Then $r_T \geq r_d := 1/(11.15(20\mu + 1))$ and $n^* \leq N$. The prefactor in Lemma 5.1 is monotone in (r_T, n^*) , so we may substitute (r_d, N) . Lemma 3.1 gives $V^2 \leq 81\mu^6 \|h\|^2$, so the high-zero term is at most

$$H(\mu) \|h\|^2, \quad H(\mu) = 2\sqrt{\mu} 81\mu^6 (20\mu + 1) / (\pi e^{20\mu}) \leq e^{-16\mu}.$$

This proves Theorem C with

$$(6.1) \quad P(\mu) = \frac{2N}{r_d^2} \left(\frac{1}{2} + \frac{\sqrt{\mu}}{2} \log \frac{1}{2r_d} \right) (592 + 152.2K_1^2),$$

where

$$r_d = \frac{1}{11.15(20\mu + 1)}, \quad N = \frac{20\mu + 1 - \log 2\pi}{4\pi} + 2R(20\mu + 1), \quad K_1 = 2 \log(c/3) + 8.08.$$

Ball arithmetic gives the values $P(5) \leq 1.854 \times 10^{13}$, $P(15) \leq 1.151 \times 10^{15}$ and $P(100) \leq 1.557 \times 10^{18}$. Each factor of P , divided by its power of μ or $\log \mu$, is non-increasing, so $P(\mu)/(\mu^{7/2}(\log \mu)^3)$ is non-increasing for $\mu \geq 5$. $\square$

7. THE HERMITE LIMIT AND THEOREM D

7.1. Rescaled operator. In the variable $x = \lambda s$, the operator L_c becomes μA_μ , where

$$A_\mu = -\frac{d}{dx} \left(1 - \frac{x^2}{\mu} \right) \frac{d}{dx} + 4\pi^2 x^2 \quad \text{on } (-\lambda, \lambda),$$

with the natural boundary condition. The function $h_{n,\lambda}$ is its n -th eigenfunction, with eigenvalue $E_n(\mu) = \chi_n^{\text{SL}}/\mu$. The limit operator is the harmonic oscillator $A_\infty = -d^2/dx^2 + 4\pi^2 x^2$. Its eigenfunctions are $g_n(x) = (2\pi)^{1/4} \tilde{H}_n(\sqrt{2\pi}x)$, with eigenvalues $E_n^\infty = 2\pi(2n+1)$, and $\hat{g}_n = g_n$ for $n \in \{0, 4\}$. Put

$$h_\infty = I_4^\infty g_0 - I_0^\infty g_4, \quad I_n^\infty = g_n(0), \quad e_\infty(y) = e^{y/2} \sum_{m \geq 1} h_\infty(m e^y),$$

and $\kappa_\infty = \|e_\infty\|^2 / \|h_\infty\|^2$. Then $h_\infty(0) = \int h_\infty = 0$, and by Poisson summation e_∞ is even. Ball arithmetic gives $0.219247199548 < \kappa_\infty < 0.219247199550$.

7.2. Residuals. On $(-\lambda, \lambda)$ we have $(A_\mu - E_n^\infty)g_n = \mu^{-1}(x^2 g_n')'$. After scaling,

$$\|(x^2 g_n')'\|^2 = \varrho_n^2 := \int_{\mathbb{R}} ((u^2 \tilde{H}_n')')^2 du,$$

with $\varrho_0^2 = \frac{33}{16}$, $\varrho_2^2 = \frac{585}{16}$ and $\varrho_4^2 = \frac{3553}{16}$. Let $\tau_n = \int_{|u| > \sqrt{c}} \tilde{H}_n^2$ and $s_n = \sqrt{1 - \tau_n}$. The normalized restriction $g_n^\sharp = g_n|_{(-\lambda, \lambda)}/s_n$ then has residual $\|(A_\mu - E_n^\infty)g_n^\sharp\| \leq \delta_n = \varrho_n/(\mu s_n)$.

7.3. Localization of the spectrum.

- **Each interval contains an eigenvalue.** Each interval $[E_{2k}^\infty - \delta_{2k}, E_{2k}^\infty + \delta_{2k}]$, $k = 0, 1, 2$, contains an eigenvalue.
- **Lower bound for the fourth even eigenvalue.** IMS localization uses the ramp cut-off j_1, j_2 , which has $|j_1'|^2 + |j_2'|^2 \leq \pi^2/R^2$, with $R^2 = \theta\mu$. It gives

$$\varepsilon_4 \geq L_6 := \min\{(1 - \theta)E_6^\infty, \pi^2\theta\mu\} - \frac{\pi^2}{\theta\mu}.$$

- **Counting.** If the three intervals are disjoint and $L_6 > E_4^\infty + \delta_4$, then $E_{2k}(\mu)$ lies in the k -th interval.

This yields the lower bounds

$$\gamma_0 = E_2^\infty - \delta_2 - E_0^\infty, \quad \gamma_4 = \min\{E_4^\infty - E_2^\infty - \delta_2, L_6 - E_4^\infty\}$$

for the distance from E_0^∞, E_4^∞ to the rest of the even spectrum. The odd sector does not interact with even trial functions.

7.4. Davis–Kahan in the form norm. Let $q_\mu(f) = \int_{-\lambda}^\lambda (1 - x^2/\mu)|f'|^2 + 4\pi^2 x^2|f|^2$ be the form of A_μ . Let φ be the normalized eigenfunction for $E_n(\mu)$, with the sign chosen so that $\langle g_n^\sharp, \varphi \rangle \geq 0$, and put $u = g_n^\sharp - \langle g_n^\sharp, \varphi \rangle \varphi$. Since $(A_\mu - E_n^\infty)u = P^\perp(A_\mu - E_n^\infty)g_n^\sharp$,

$$\|u\| \leq a_n := \frac{\delta_n}{\gamma_n}, \quad q_\mu(u) = \langle (A_\mu - E_n^\infty)u, u \rangle + E_n^\infty \|u\|^2 \leq \frac{\delta_n^2}{\gamma_n} + \frac{E_n^\infty \delta_n^2}{\gamma_n^2},$$

and $4\pi^2 \|xu\|^2 \leq q_\mu(u)$. Put $b_n = \sqrt{\delta_n^2/\gamma_n + E_n^\infty \delta_n^2/\gamma_n^2}/(2\pi)$. Let $h_{n,\lambda} = \varphi$, extended by zero, and $v_n = h_{n,\lambda} - g_n$ on $\mathbb{R}$. Using $1 - \langle g_n^\sharp, \varphi \rangle \leq a_n^2$, $\|x\varphi\| \leq \sqrt{E_n^\infty + \delta_n}/(2\pi)$ and $\|xg_n\|^2 = (2n+1)/(4\pi)$, we get

$$\begin{aligned} \|v_n\| &\leq V_n := a_n + a_n^2 + (1 - s_n) + \sqrt{\tau_n}, \\ \|xv_n\| &\leq X_n := b_n + a_n^2 \frac{\sqrt{E_n^\infty + \delta_n}}{2\pi} + (s_n^{-1} - 1) \sqrt{\frac{2n+1}{4\pi}} + \|xg_n \mathbf{1}_{|x|>\lambda}\|. \end{aligned}$$

7.5. Lattice sums. For $(Tv)(y) = e^{y/2} \sum_m v(me^y)$, $Y_0 = 0.8$ and even v :

$$\|Tv\|_{L^2(|y|\leq Y_0)}^2 \leq \frac{\pi^2}{12} e^{Y_0} W_{Y_0}(v)^2, \quad W_{Y_0}(v)^2 = \int_{\mathbb{R}} (2 \sinh(Y_0)x^2 + |x|)v^2.$$

(Cauchy–Schwarz with weights m^{-2} , the substitution $x = me^y$, and the count $\#\{m : x \in [me^{-Y_0}, me^{Y_0}]\} \leq 2x \sinh Y_0 + 1$.) Since $|x| \leq (1 + x^2)/2$,

$$W_{Y_0}(v_n) \leq B_n := \sqrt{(2 \sinh Y_0 + \frac{1}{2})X_n^2 + \frac{1}{2}V_n^2}, \quad W_{Y_0}(h_{n,\lambda}) \leq H_n := W_{Y_0}(g_n) + B_n,$$

and, since $\|(1 + x^2)^{-1/2}\|^2 = \pi$, $|I_n(\lambda) - I_n^\infty| \leq D_n := \sqrt{\pi} \sqrt{V_n^2 + X_n^2}$.

7.6. Conclusion. For $\mu \geq 5$, the window contains $[-Y_0, Y_0]$. Hence

$$\|k_\lambda\| \geq \|Th_\infty\|_{Y_0} - \|T(h_\lambda - h_\infty)\|_{Y_0}, \quad \|Th_\infty\|_{Y_0} \geq 0.65294,$$

where the tail of e_∞ outside $|y| \leq 0.8$ is below 10^{-8} . With $\alpha_\infty = I_4^\infty$, $\beta_\infty = -I_0^\infty$ and $h_\lambda - h_\infty = (\alpha_\lambda - \alpha_\infty)h_{0,\lambda} + \alpha_\infty v_0 + (\beta_\lambda - \beta_\infty)h_{4,\lambda} + \beta_\infty v_4$,

$$\|T(h_\lambda - h_\infty)\|_{Y_0} \leq U_T := \sqrt{\frac{\pi^2}{12} e^{Y_0}} [D_4 H_0 + |\alpha_\infty| B_0 + D_0 H_4 + |\beta_\infty| B_4],$$

$$\|h_\lambda\|^2 \leq D_H := (|\alpha_\infty| + D_4)^2 + (|\beta_\infty| + D_0)^2,$$

so that, whenever $0.65294 - U_T > 0$ and the separation conditions above hold,

$$\frac{\|k_\lambda\|^2}{\|h_\lambda\|^2} \geq F_*(\mu, \theta) := \frac{(0.65294 - U_T)^2}{D_H}.$$

For fixed θ , δ_n and the tails decrease in μ while L_6 and the γ_n increase, so U_T and D_H decrease and F_* increases. Ball arithmetic gives:

(μ, θ)	$\gamma_4 \geq$	$V_4 \leq$	$X_4 \leq$	$U_T \leq$	$F_* \geq$
(50, 0.145)	11.92	0.0257	0.0337	0.178	0.1092
(80, 0.095)	16.07	0.0118	0.0159	0.0844	0.1615

This proves Theorem D. For the $\liminf$, run the same argument with an arbitrary fixed Y in place of Y_0 : the weighted errors tend to 0, so $\liminf \|k_\lambda\|^2 \geq \|Th_\infty\|_{[-Y, Y]}^2$, and $\|h_\lambda\|^2 \rightarrow \|h_\infty\|^2$; then let $Y \rightarrow \infty$. $\square$

8. PROOF OF THEOREM E

By Theorem C, Lemma 2.1 and [4, Thm. 1],

$$|W(k_\lambda)| \leq (P(\mu)C_4(2\pi\mu)^{9/2}e^{-4\pi\mu} + e^{-16\mu})\|h_\lambda\|^2,$$

where $C_4 \leq 6621$ [4, Thm. 1 and Table 1]. Since $\lambda_{\min} \leq W(k_\lambda)/\|k_\lambda\|^2$ [5, (2.1)], Theorem D gives the first bound. The function $P(\mu)/(\mu^{7/2}\log^3\mu)$ and the term $e^{-16\mu+4\pi\mu}/(\mu^8\log^3\mu)$ are non-increasing, so their values at $\mu = 50$ give the constant 5.011×10^{17} . $\square$

9. NUMERICAL EVIDENCE

μ	5	7	11	13	15
$W(k_\lambda)/(\rho_{\text{out}}\ h\ ^2)$	1.62	1.80	2.05	2.12	2.25
$\ k_\lambda\ ^2/\ h_\lambda\ ^2$	0.2227	0.2214	0.2205	0.2203	0.2201
ρ_{out}	1.9×10^{-18}	1.1×10^{-28}	1.4×10^{-49}	3.6×10^{-60}	8.5×10^{-71}

TABLE 1. Galerkin computations (not certified). The value of $\|k_\lambda\|^2/\|h_\lambda\|^2$ at $\mu = 50$ is 0.2195.

APPENDIX A. PROOF OF LEMMA 3.1

Hermite functions. Let $\mathcal{F}_1 g(\omega) = (2\pi)^{-1/2} \int e^{-i\omega s} g(s) ds$. Let P be multiplication by $\mathbf{1}_{[-1,1]}$ and $B_c = \mathcal{F}_1^{-1} \mathbf{1}_{[-c,c]} \mathcal{F}_1$. For f on $[-1,1]$ extended by zero, $\langle Q_c f, f \rangle = \|B_c f\|^2$.

Let $u_k(x) = H_k(x)e^{-x^2/2}/(\pi^{1/4}\sqrt{2^k k!})$ and $\phi_k(s) = c^{1/4}u_k(\sqrt{c}s)$ for $0 \leq k \leq 4$. These are orthonormal, and $\mathcal{F}_1 u_k = (-i)^k u_k$. For g in $G = \text{span}(\phi_0, \dots, \phi_4)$, Cauchy–Schwarz gives

$$\|(1-P)g\|^2 \leq \varepsilon(c)\|g\|^2, \quad \|(1-B_c)g\|^2 \leq \varepsilon(c)\|g\|^2, \quad \varepsilon(c) = \sum_{k \leq 4} \int_{|x| > \sqrt{c}} u_k^2.$$

The leakage. We have $\sum_{k \leq 4} H_k(x)^2/(2^k k!) = \frac{2}{3}x^8 - \frac{8}{3}x^6 + 5x^4 + \frac{15}{8} \leq \frac{181}{24}x^8$ for $x \geq 1$. Integration by parts gives $\int_R^\infty x^8 e^{-x^2} \leq e^{-R^2} R^7$ for $R^2 \geq 30$. Hence

$$\varepsilon(c) \leq \frac{181}{12} e^{-c} c^{7/2} \leq 16e^{-c} c^4 \leq 16e^{-30} 30^4 \leq \varepsilon_* = \frac{16 \cdot 14!}{30^{10}} < \frac{1}{400},$$

using $e^{30} \geq 30^{14}/14!$.

The eigenvalue bound. Let $U = PG$, which has dimension 5. For $f = Pg \in U$, $\|(1-B_c)f\| \leq 2\sqrt{\varepsilon_*}\|g\|$ and $\|f\|^2 \geq (1-\varepsilon_*)\|g\|^2$. Hence $\langle Q_c f, f \rangle / \|f\|^2 \geq 1 - 4\varepsilon_*/(1-\varepsilon_*)$. Since U contains a vector orthogonal to $\psi_0, \dots, \psi_3$, we get $\lambda_4 \geq (1-5\varepsilon_*)/(1-\varepsilon_*) > 395/399$.

Pointwise bounds. From $\psi_n = \mu_n^{-1} \mathcal{F}_c \psi_n$ and $|\mu_n|^2 = 2\pi\lambda_n/c$:

$$|\psi_n| \leq \sqrt{2}/|\mu_n| = \sqrt{c/(\pi\lambda_n)}, \quad |\psi'_n| \leq c\sqrt{2/3}/|\mu_n| = c^{3/2}/\sqrt{3\pi\lambda_n}.$$

Variation. Let $H = \|h\|$. Since $|\alpha| + |\beta| \leq \sqrt{2}H$,

$$M = \sup |h| \leq 2\sqrt{2}\mu^{1/4}H, \quad D = \sup |h'| \leq 4\pi\sqrt{2/3}\mu^{3/4}H.$$

At most $N \leq \mu$ terms $k_n(y) = e^{y/2}h(ne^y)$ meet the window. Each has smooth variation at most $\sqrt{\lambda}M + \lambda^{3/2}D$, and its jumps at the window endpoints and at its cut point total at most $(\sqrt{\lambda} + \lambda^{-1/2})M$. Hence

$$V \leq \mu(3\sqrt{\lambda}M + \lambda^{3/2}D) \leq (6\sqrt{2}\mu^{3/2} + 4\pi\sqrt{2/3}\mu^{5/2})H \leq 9\mu^3H.$$

APPENDIX B. PROOF OF LEMMA 3.3

Let $s = \chi_n^{\text{SL}}/c^2 \in [0, 0.3]$, $p = t^2 - 1$, $r = t^2 - s$, $q = c^2 r$ and $A_0 = \psi_n(1) \geq 0$ (after a sign change).

Near the endpoint. By Lemma 3.2(b), $|\tilde{\psi}| \leq A_0$. Integrating the ODE gives $|p\tilde{\psi}'(t)| \leq A_0 c^2(t-1)t^2$. With $t_0 = 1 + 1/(2c^2) < 1801/1800$ this gives

$$|\tilde{\psi}(t_0) - A_0| \leq A_0 t_0^2/4, \quad \text{hence} \quad \tilde{\psi}(t_0) \geq \frac{37}{50} A_0.$$

Liouville–Green error. Write $D = s^2 + 2s + (3 - 6s)t^2$ as in Lemma 3.2(c). Then $0 \leq D \leq 3t^2$, since $D - 3t^2 = s(s + 2 - 6t^2) \leq 0$, and $r \geq \frac{7}{10}t^2$. Integrating the two positive parts of δ separately gives

$$J = \int_{t_0}^{\infty} |\delta| z' dt \leq \frac{3}{10} + \frac{9}{196} = \frac{339}{980} < \frac{7}{20}.$$

For the first part we use $c\sqrt{t_0^2 - 1} \geq 1$. For the second we use $\int_1^{\infty} dt/(t^3\sqrt{t^2 - 1}) = \pi/4$.

Energy. For $E = w^2 + w_z^2$ we have $|E_z| \leq |\delta|E$, so $\frac{7}{10}E_0 \leq E \leq \frac{10}{7}E_0$ on $[t_0, \infty)$. Moreover

$$E_0 \geq w(t_0)^2 = \sqrt{pq} \tilde{\psi}(t_0)^2 \geq \frac{5}{6} \left(\frac{37}{50}\right)^2 A_0^2.$$

Mass. On $t \geq 4/3$, put $N = 4c^2 pr^3 \delta = r^2 - pD$. With $v = t^2 - 1$ one has $2r^2 + N = 6sv^2 + (3 - 2s - s^2)v + 3(1 - s)^2 \geq 0$ and $2r^2 - N = (4 - 6s)v^2 + (5 - 6s + s^2)v + (1 - s)^2 \geq 0$, so $|N| \leq 2r^2$ and $|\delta| \leq 1/(2c^2 pr) < 1/2000$. Put $g = 1/q$, a decreasing function of z . The identity $E = (2 + \delta)w^2 + (ww_z)_z$, multiplied by g and integrated by parts, gives

$$T := \int_{4/3}^{\infty} \tilde{\psi}^2 \geq \frac{E_0}{2 + 1/2000} \left(\frac{7}{10} I - \frac{10}{7q(4/3)} \right), \quad I = \int_{4/3}^{\infty} \frac{dt}{\sqrt{pq}} \geq \frac{\arcsin(3/4)}{c} > \frac{21}{25c}.$$

Since $\tilde{\psi}_n$ has parity $(-1)^n$, $\int_{|t|>1} \tilde{\psi}^2 \geq 2T \geq C_* A_0^2/c$ with $C_* = 118059822/465616375 > \frac{1}{4}$. The mass identity $\int_{|t|>1} \tilde{\psi}^2 = (1 - \lambda_n)/\lambda_n$ completes the proof.

REPRODUCIBILITY

All computer-assisted steps are reproducible from the public repository ([URL and DOI to be added on release](#)), which contains every script, its raw output, and a table matching each constant of this paper to its script. The constants of Lemmas 3.2, 3.3, 3.5, 3.6 and 4.1, of $P(\mu)$, of κ_{∞} , of Theorem D and of Theorem E are computed in ball arithmetic (Arb, via python-flint) with directed rounding asserted in the scripts. The Galerkin values in Section 9 are not part of any proof; the script records the parameters needed for convergence.

USE OF GENERATIVE AI

This work was carried out with extensive use of generative AI systems, namely Anthropic Claude, Google Gemini and OpenAI GPT. Under the direction of the author, these systems drafted the proofs, implemented and ran the computations, reviewed one another's drafts adversarially, searched the literature, and drafted this manuscript. The author chose the problem and the strategy, assigned and reviewed the tasks, and takes full responsibility for the content. The AI systems are not authors of this paper.

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